

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular Winter Examination – 2024

Course: BTech Branch : Electronics and Computer Engineering/ Electronics and Computer Science

Semester :VII

Subject Code & Name: BTECPC702 DEEP LEARNING

Max Marks: 60

Date:07/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	What is the main component of a simple artificial neural network? a) Neurons b) Layers c) Weights d) All of the above	(L1/CO1)	1
2	Which of the following is NOT an activation function used in CNNs? a) Sigmoid b) ReLU c) Tanh d) Convolution	(L1/CO2)	1
3	High bias indicates a) Overfitting b) Underfitting c) High variance d) Low variance	(L1/CO3)	1
4	The gates in LSTM include: a) Update and reset gates b) Input, forget, and output gates c) Weight and bias gates d) Memory gates only	(L1/CO4)	1
5	Image denoising is a task where the model: a) Adds noise to images b) Removes noise from images c) Detects objects in images d) Segments objects in images	(L1/CO5)	1
6	What function do activation functions typically perform in a neural network? a) Scale the inputs b) Add non-linearity c) Update weights d) Optimize gradients	(L1/CO1)	1
7	What is the purpose of a filter (or kernel) in a convolutional layer? a) Extract features from the input b) Pool data c) Perform activation d) Reduce dimensionality	(L1/CO2)	1
8	Which domain commonly uses transfer learning? a) Image recognition b) Natural language processing c) Speech recognition d) All of the above	(L1/CO3)	1
9	Which of the following is a sequential model? a) CNN b) RNN c) Decision Tree d) SVM	(L1/CO4)	1
10	A GAN consists of two main components: a) Encoder and decoder b) Generator and discriminator c) Classifier and segmenter d) Autoencoder and transformer	(L1/CO5)	1

11	Which of the following is NOT an optimization algorithm? a) Adam b) RMSProp c) Gradient Descent d) ReLU	(L1/CO1)	1
12	Which activation function is most commonly used in CNNs? a) Sigmoid b) Tanh c) ReLU d) Softmax	(L2/CO2)	1
Q. 2	Solve the following.		12
A)	Illustrate the working principle of Simple Artificial Neural Network With suitable diagram.	(L2/CO1)	6
B)	Discuss Forward Propagation and Backward Propagation Learning in Deep Neural Network.	(L2/CO1)	6
Q.3	Solve the following.		12
A)	Explain building blocks of CNN with suitable labelled diagram.	(L2/CO2)	6
B)	Discuss Different CNN Architectures in detail.	(L2/CO2)	6
Q. 4	Solve Any Two of the following.		12
A)	Define the Transfer Learning and identify the need of Transfer Learning in Deep Neural Network.	(L1,L2/CO3)	6
B)	Differentiate Between Early Stopping and Drop Out in Deep Neural Network.	(L1/CO3)	6
C)	Illustrate Data Augmentation in Deep Neural Network.	(L2/CO3)	6
Q.5	Solve Any Two of the following.		12
A)	Illustrate the working of Recurrent Neural Network with neat labelled diagram.	(L2/CO4)	6
B)	Summarize the working principle of LSTM with neat labelled diagram.	(L2/CO4)	6
C)	Outline the working concept of GRU-Gated Recurrent Unit with suitable diagram.	(L2/CO4)	6
Q. 6	Solve Any Two of the following.		12
A)	Elaborate Neural Style Transfer Network in detail.	(L2/CO5)	6
B)	Discuss Variational Auto Encoder.	(L1/CO5)	6
C)	Write Short note on Generative Adversarial Networks	(L1/CO5)	6
	*** End ***		